

COSC 594/650

Gradient Descent for Convex Objectives

Dongsheng Ding

dongshed@utk.edu

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

Convexity vs strong convexity

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (differentiable)

■ CONVEX FUNCTION

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

★ $\nabla^2 f(x) \succeq 0$ – PSD

■ STRONGLY CONVEX FUNCTION

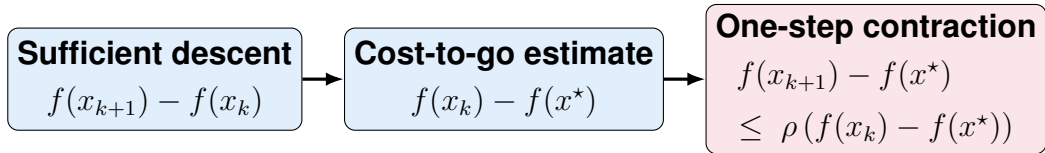
$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2$$

★ $\nabla^2 f(x) \succeq \mu I$ – PD

★ $\mu I \preceq \nabla^2 f(x) \preceq LI$ – PD and Lipschitz gradient (L)

★ $\kappa = \frac{L}{\mu}$ – condition number

Proof machinery



Iterate the contraction:

$$e_{k+1} \leq \rho e_k \implies e_k \leq \rho^k e_0$$

Convergence analysis for general convex functions

Gradient descent: $x_{k+1} = x_k - \alpha \nabla f(x_k)$, $k = 0, 1, 2, \dots$

■ DESCENT LEMMA

$$f(x_{k+1}) \leq f(x_k) - \frac{2\alpha - \alpha^2 L}{2} \|\nabla f(x_k)\|^2$$

$$\implies \sum_{i=0}^k \|\nabla f(x_i)\|^2 \leq \frac{2}{2\alpha - \alpha^2 L} (f(x_0) - f(x^*)) \leq \frac{L \|x_0 - x^*\|^2}{2\alpha - \alpha^2 L}$$

■ CONVEXITY

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &= \|x_k - x^*\|^2 - 2\alpha \langle \nabla f(x_k), x_k - x^* \rangle + \alpha^2 \|\nabla f(x_k)\|^2 \\ &\leq \|x_k - x^*\|^2 - 2\alpha (f(x_k) - f(x^*)) + \alpha^2 \|\nabla f(x_k)\|^2 \end{aligned}$$

$$\implies \sum_{i=0}^k (f(x_i) - f(x^*)) \leq \frac{1}{2\alpha} \left(\|x_0 - x^*\|^2 + \frac{\alpha^2 L \|x_0 - x^*\|^2}{2\alpha - \alpha^2 L} \right)$$

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$$\implies f(x_{k+1}) - f(x^*) \leq f(x_k) - f(x^*)$$

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$$\implies f(x_k) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{k+1}, \quad \alpha = \frac{1}{L}$$

Iteration complexity for general convex functions

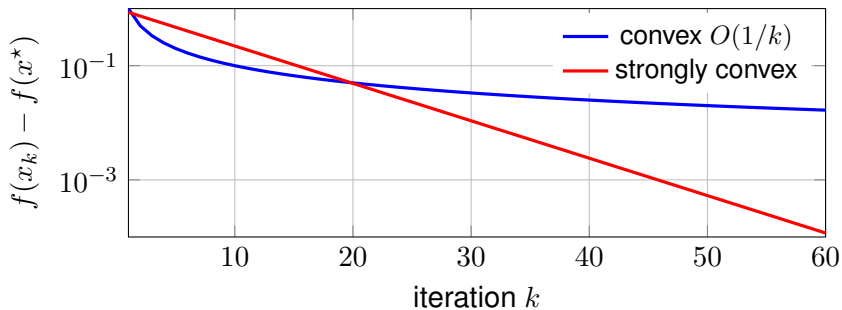
$$e(x_k) \leq \frac{L\|x_0 - x^\star\|^2}{k+1} \leq \epsilon$$

$$\implies k \geq \frac{L\|x_0 - x^\star\|^2}{\epsilon} - 1$$

Iteration complexity: $\mathcal{O}\left(\frac{L\|x_0 - x^\star\|^2}{\epsilon}\right)$

Sublinear rate

Rates at a glance



QUESTION: Is the dependence on k and κ improvable?

First-order oracle model and complexity

Function class $\mathcal{F}(D, L)$: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (convex)

Initial gap: $\|x_0 - x^*\| \leq D$

Lipschitz gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$

■ ORACLE MODEL Ω OF FIRST-ORDER ALGORITHMS

$$x_{k+1} = \sum_{i=0}^k (\alpha_{i,k} x_i + \beta_{i,k} \nabla f(x_i))$$

★ $\alpha_k = (\alpha_{0,k}, \alpha_{1,k}, \dots, \alpha_{k,k}), \beta_k = (\beta_{0,k}, \beta_{1,k}, \dots, \beta_{k,k})$ – weights

■ COMPLEXITY OF AN ALGORITHM $\mathcal{A} \in \Omega$

$$\mathcal{C}(\mathcal{A}, \epsilon) = \sup_{f \in \mathcal{F}(D, L)} \min\{k \mid f(x_k) - f(x^*) \leq \epsilon\}$$

Optimal first-order methods

Optimal complexity: $\inf_{\mathcal{A} \in \Omega} \mathcal{C}(\mathcal{A}, \epsilon)$

Iteration complexity:

$$\sqrt{\frac{3}{32}} \min \left(n, \sqrt{\frac{LD^2}{\epsilon}} \right) \leq \inf_{\mathcal{A} \in \Omega} \mathcal{C}(\mathcal{A}, \epsilon) \leq \sqrt{\frac{4LD^2}{\epsilon}}$$

Thm 3.14, Bubeck Nesterov 1983

Gradient descent is not order optimal

Questions?

(**Email: dongshed@utk.edu**)